

Assignment 3 - Practice: Classify the System

For each augmented matrix, the original problem asks whether the system is consistent and whether it has a unique, infinite, or no solution. Each question is followed by its answer.

Q1

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 1 & | & 2 \\ 0 & 1 & | & 1 \end{bmatrix}$$

Answer: 2 variables; 2 pivots, so $\text{rank}(A)=\text{rank}([A|b])=2$. The final row is not contradictory. **Consistent; unique solution.**

Back-substitution gives $y=1$ and $x=1$, so $(x,y)=(1,1)$.

Q2

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 2 & | & 4 \\ 2 & 4 & | & 8 \end{bmatrix}$$

Answer: 2 variables; the second row is twice the first, so $\text{rank}(A)=\text{rank}([A|b])=1$. No contradictory row; one free variable. **Consistent; infinitely many solutions.**

Let $y=t$. Then $x=4-2t$, so $(x,y)=(4-2t,t)$.

Q3

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 1 & | & 2 \\ 1 & 1 & | & 5 \end{bmatrix}$$

Answer: 2 variables. Subtract row 1 from row 2 to get $[0 \ 0 \ | \ 3]$, which is a contradiction. Thus $\text{rank}(A)=1$ but $\text{rank}([A|b])=2$. **Inconsistent; no solution.**

Q4

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 0 & 2 & | & 3 \\ 0 & 1 & 1 & | & 2 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

Answer: 3 variables; 3 pivots and $\text{rank}(A)=\text{rank}([A|b])=3$. The last row gives $z=1$, then $y=1$ and $x=1$. **Consistent; unique solution $(x,y,z)=(1,1,1)$.**

Q5

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 2 & 4 & 6 & | & 12 \\ 1 & 1 & 1 & | & 3 \end{bmatrix}$$

Answer: 3 variables. R_2-2R_1 gives a zero row; R_3-R_1 gives $[0 \ -1 \ -2 \ | \ -3]$. Hence $\text{rank}(A)=\text{rank}([A|b])=2$, with one free variable and no contradiction. **Consistent; infinitely many solutions.**

Let $z=t$. Then $y=3-2t$ and $x=t$, so $(x,y,z)=(t,3-2t,t)$.

Q6

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 2 & 4 & 6 & | & 10 \\ 1 & 1 & 1 & | & 3 \end{bmatrix}$$

Answer: 3 variables. R_2-2R_1 gives $[0 \ 0 \ 0 \ | \ -2]$, a contradiction. Thus $\text{rank}(A)=2$ but $\text{rank}([A|b])=3$. **Inconsistent; no solution.**

Q7

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Answer: 3 variables; 2 pivots, so $\text{rank}(A)=\text{rank}([A|b])=2$. The last row is $0=0$; one variable is free. **Consistent; infinitely many solutions.**

Let $z=t$. Then $x=2-t$ and $y=1+t$, so $(x,y,z)=(2-t,1+t,t)$.

Q8

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & 0 & | & 5 \end{bmatrix}$$

Answer: 3 variables; the last row states $0=5$, a contradiction. Thus $\text{rank}(A)=2$ but $\text{rank}([A|b])=3$. **Inconsistent; no solution.**

Q9

Question: Classify the system represented by this augmented matrix:

$$\begin{bmatrix} 1 & 2 & 0 & | & 3 \\ 0 & 1 & 4 & | & 2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Answer: 3 variables; 2 pivots, so $\text{rank}(A)=\text{rank}([A|b])=2$. The last row is $0=0$; there is one free variable. **Consistent; infinitely many solutions.**

Let $z=t$. Then $y=2-4t$ and $x=-1+8t$, so $(x,y,z)=(-1+8t,2-4t,t)$.